

Cluster 10701

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Subjects

Research fields: Biology, Medicine, Computer science

The most common research fields of articles in this cluster. [Read more.](#)

Research subfields: Oncology, Cancer research, Bioinformatics

Common research subfields of articles in this cluster. [Read more.](#)

Key concepts: breast cancer, gene expression, triple-negative breast cancer, cancer patients, Breast Cancer International

These key concepts are identified algorithmically using article titles and abstracts. [Read more.](#)

15.81% of all articles are AI-related

Articles are algorithmically labeled as AI-related or not. The percentage is based on articles in the cluster from the five year period ending in 2023 with English-language abstracts. [Read more.](#)

0.00% of English-language articles are related to robotics

Articles are labeled as robotics-related or not using an ETO model. The percentage is based on articles in the cluster from the five year period ending in 2023 with English-language abstracts. [Read more.](#)

0.69% of English-language articles are related to computer vision

Articles are labeled as computer vision-related or not using an ETO model. The percentage is based on articles from the five year period ending in 2023 with English-language abstracts. [Read more.](#)

0.34% of English-language articles are related to natural language processing

Articles are labeled as NLP-related or not using an ETO model. The percentage is based on articles from the five year period ending in 2023 with English-language abstracts. [Read more.](#)

0.00% of English-language articles are related to AI safety

Articles are labeled as related to AI safety or not using an ETO model. The percentage is based on articles from the five year period ending in 2023 with English-language abstracts. [Read more.](#)

0.00% of English-language articles are related to cybersecurity

Vital signs

298 recent articles in the cluster

298 articles in this cluster were published in the five year period ending in 2023.

Average article is **13.35** years old

The average age of articles in this cluster. However, most Map of Science metrics are calculated using articles from the past five years only.

Growth rating: **58.60th** percentile

Over the three year period ending in 2023, this cluster grew faster than roughly 58.60% of other clusters.

Citation rating: **42.60th** percentile

A cluster's citation rating is equal to the average citation percentile for all the articles in the cluster. Citation percentiles are calculated relative to articles of the same publication year. In this case, the average article in this cluster is cited more often than roughly 42.60% of all other articles published in the same year.

90.60% of articles are in English

Not expected to grow extremely fast

As determined by ETO's growth prediction model. [Read more.](#)

Countries and languages

Top author countries

Each row lists the number of articles in this cluster from the last five years with at least one author from an organization in the specified country. Data available for 227 of 298 articles. 172 of these articles had at least one author from an organization in these ten countries.

Country	Articles (five year period ending in 2023)	Yearly citations per article (average)
United States	71	1.76
China	39	1.36
India	20	1.37
United Kingdom	14	2.44
France	13	4.63
Japan	11	1.17
Canada	10	3.19
South Korea	9	1.09
Italy	9	2.61
Taiwan	8	1.53

Top funding countries

Data available for 100 of 298 articles. 82 of these articles disclosed funding from one or more organizations in these ten countries.

Country	Articles (last five years)	Yearly citations (average)
United States	25	2.26
China	23	1.45
Canada	7	4.28
South Korea	7	0.97
United Kingdom	5	7.36
Taiwan	5	1.60
Sweden	4	3.21
Spain	4	6.95
Japan	4	0.90
Belgium	3	6.17

Articles and sources

Core articles

Core articles are especially highly connected to other papers within the cluster.

[An Improved, Assay Platform Agnostic, Absolute Single Sample Breast Cancer Subtype Classifier](#)

2020: Cancers. 10 citations.

[RNA sequencing-based single sample predictors of molecular subtype and risk of recurrence for clinical assessment of early-stage breast cancer](#)

2021: npj Breast Cancer. 45 citations.

[Machine learning for RNA sequencing-based intrinsic subtyping of breast cancer](#)

2020: Scientific reports. 46 citations.

[A single nucleotide variant on chromosome 9 residing within NFIB distinguishes patients with basal-like human breast cancer.](#)

2022. 16 citations.

[Prognostic gene expression signatures of breast cancer are lacking a sensible biological meaning](#)

2021: Scientific reports. 34 citations.

[A single nucleotide variant on chromosome 12 residing within RAB3IP distinguishes patients with luminal B breast cancer.](#)

2022. 2 citations.

[Residual risk stratification of Taiwanese breast cancers following curative therapies with the extended concurrent genes signature](#)

2021: Breast cancer research and treatment. 5 citations.

Review articles

Review articles have between 100 and 1000 out-citations. These articles describe and systematize other contributors' research.

[A systematic review of datasets that can help elucidate relationships among gene expression, race, and immunohistochemistry-defined subtypes in breast cancer](#)

2021: Cancer biology & therapy. 4 citations.

[Annotated Compendium of 102 Breast Cancer Gene-Expression Datasets](#)

2023: bioRxiv (Cold Spring Harbor Laboratory). 1 citations.

Highly cited articles

Highly cited articles are the articles in the cluster with the most citations overall.

[RNA sequencing-based single sample predictors of molecular subtype and risk of recurrence for clinical assessment of early-stage breast cancer](#)

2021: npj Breast Cancer. 45 citations.

[Defining human mesenchymal and epithelial heterogeneity in response to oral inflammatory disease](#)

2020: eLife. 71 citations.

[Machine learning for RNA sequencing-based intrinsic subtyping of breast cancer](#)

2020: Scientific reports. 46 citations.

[Molecular intrinsic versus clinical subtyping in breast cancer: A comprehensive review](#)

2020: Clinical genetics. 69 citations.

[New Advances in Molecular Breast Cancer Pathology.](#)

2020: Seminars in cancer biology. 63 citations.

[Comprehensive characterization of claudin-low breast tumors reflects the impact of the cell-of-origin on cancer evolution](#)

2020: Nature communications. 72 citations.

[bc-GenExMiner 4.5: new mining module computes breast cancer differential gene expression analyses](#)

2021: Database : the journal of biological databases and curation. 123 citations.

Top sources

The most common sources of articles from the last five years in this cluster, such as specific journals, conference proceedings, or preprint servers. Data available for 222 out of 298 articles. 68 of these articles came from these sources.

Source	Articles (last five years)
bioRxiv (Cold Spring Harbor Laboratory)	15
Scientific reports	14
Cancers	9
Research Square (Research Square)	8
npj Breast Cancer	5
Frontiers in immunology	5
Journal of personalized medicine	3
Cancer medicine	3
PLoS computational biology	3
IEEE Access	3

Authors, organizations, and funders

Top authors

The authors with the most articles from the last five years in this cluster. Data available for 298 of 298 articles. 44 of these articles were written by these authors.

Author	Articles (last five years)	Yearly citations (average)
Shahan Mamoor	26	0.26
Christos Sotiriou	7	0.29
Chi-Cheng Huang	5	3.28
Sherene Loi	5	8.50
Ling-Ming Tseng	5	3.28
Chi-Cheng Huang	5	6.83
Ling-Ming Tseng	5	6.73
Heinz Koelbl	5	2.40
Ching-Shui Huang	4	5.85
Tzu-Pin Lu	4	2.88

Top author organizations

The author organizations with the most articles in this cluster. Each row lists the number of articles in this cluster from the last five years with at least one author from the specified organization. Data available for 249 of 298 articles. 45 had at least one author from one of these ten organizations.

Organization	Articles (last five years)	Yearly citations (average)
NY 11730	17	0.37
Inserm - France	10	5.76
Centre National de la Recherche Scientifique - France	6	4.47
National Taiwan University - Taiwan	6	1.87
National Defense Medical Center - Taiwan	5	1.07
Taipei Medical University - Taiwan	5	1.07
National Yang Ming Chiao Tung University - Taiwan	5	2
Indiana University – Purdue University Indianapolis - United States	5	0.83
Taipei Veterans General Hospital - Taiwan	5	1.77
East Islip, NY 11730	5	0.09

Organization types

This table covers the 219 articles in this cluster from the five year period ending in 2023 with data about their author organization types (out of 298 articles total). [Read more](#)

Commercial	Education	Nonprofit	Government
1.47%	32.44%	22.81%	2.17%

Top industry organizations

The industry organizations with the most articles in this cluster. Each row lists the number of articles in this cluster from the last five years with at least one author from the specified organization. Data available for 16 out of 298 articles. 10 of these articles had at least one author from an organization listed below.

Organization	Articles (last five years)	Yearly citations (average)
Bio-Medical Science (South Korea)	1	1
Amgen (Switzerland)	1	5.67
Myriad Genetics - United States	1	0
Hitachi (Japan)	1	0.25
GlaxoSmithKline (Netherlands)	1	2
Insilicos (United States)	1	0.8
Tata Consultancy Services (India)	1	0
Unilever (United Kingdom)	1	14.8
Daiichi-Sankyo (Japan)	1	0.25
Oryzon Genomics (Spain)	1	13.5

Top funders

Data available for 127 out of 298 recent articles. 64 of these articles disclosed funding from one or more of these organizations.

Funding organization	Articles (last five years)	Yearly citations (average)
National Natural Science Foundation of China	20	1.65
National Institutes of Health - United States	18	1.59
Breast Cancer Research Foundation - United States	4	4.31
Natural Sciences and Engineering Research Council - Canada	4	3.36
Ministry of Science and Technology - Taiwan	4	1.40
Government of Spain	4	6.95
National Research Foundation - South Korea	4	1.42
National Cancer Institute - United States	4	1.23
European Union	3	2.83
European Research Council (ERC)	3	1.63

Funder types

This table covers the 92 articles in this cluster from the last five years with data about their funding organization types (out of 298 articles total).

Commercial	Education	Nonprofit	Government
0.00%	6.80%	25.17%	68.03%

Patents and industry

1187 patents cite articles in this cluster (**99.37th** percentile among all clusters)

15.95% of articles in the cluster are cited by patents

1.34% of articles have at least one author from industry

Top patent titles

Patent title	Same-cluster citations
GENE EXPRESSION PROFILES TO PREDICT BREAST CANCER OUTCOMES	30
INTEGRATION OF TUMOR CHARACTERISTICS WITH BREAST CANCER INDEX	28
POST-TREATMENT BREAST CANCER PROGNOSIS	27
GRADING OF BREAST CANCER	27
PREDICTING BREAST CANCER TREATMENT OUTCOME	24
TUMOR GRADING AND CANCER PROGNOSIS IN BREAST CANCER	17
ACCELERATED PROGRESSION RELAPSE TEST	17
A METHOD OF PREDICTING RISK OF RECURRENCE OF CANCER	15
METHOD FOR BREAST CANCER RECURRENCE PREDICTION UNDER ENDOCRINE TREATMENT	14

Top patent assignees

Patent assignee	Same-cluster citations
MASS GENERAL BRIGHAM	41
GEN HOSPITAL CORP	40
MYRIAD GENETICS	39
WASHINGTON UNIVERSITY IN ST. LOUIS	37
UNIVERSITY OF VIRGINIA	37
ELLIS MATTHEW	37
UNIVERSITY OF NORTH CAROLINA SYSTEM	34
UNIVERSITY OF UTAH	33
UNIV OF UTAH RES FOUNDATION	32

Connections

Top citing clusters

Cluster	Number of significant citations	CSET extracted phrases
9438	135	breast cancer, recurrence score, cancer patients, receptor-positive breast cancer, early breast
13840	51	Triple-Negative Breast Cancer, Breast Cancer, metastatic triple-negative breast, TNBC patients, immune cells
26279	35	Triple-negative breast cancer, breast cancer, TNBC cancer cells, TNBC patients, metastatic TNBC cancer
3259	29	Triple-Negative Breast Cancer, Breast Cancer, TNBC patients, metastatic triple-negative breast, TNBC cell
22059	28	immune cell infiltration, Cancer Genome Atlas, immune cells, tumor immune microenvironment, breast cancer
14251	13	cancer driver genes, Cancer Genome Atlas, cancer types, driver mutations, identify cancer driver
1526	9	single-cell RNA sequencing, cell types, spatial transcriptomics, transcriptomic cell types, RNA sequencing data
40042	9	Multi-omics Data Integration, omics data, data integration methods, integrating multi-omics data, data analysis
1276	9	Breast Cancer, pathologic complete response, Neoadjuvant Chemotherapy, triple-negative breast cancer, pCR
31279	7	Breast Cancer, cancer patients, triple-negative breast cancer, Early Breast, Cancer Working Group

Top cited clusters

Cluster	Number of significant citations	CSET extracted phrases
3259	77	Triple-Negative Breast Cancer, Breast Cancer, TNBC patients, metastatic triple-negative breast, TNBC cell
9438	53	breast cancer, recurrence score, cancer patients, receptor-positive breast cancer, early breast
13840	20	Triple-Negative Breast Cancer, Breast Cancer, metastatic triple-negative breast, TNBC patients, immune cells
2523	17	gene expression analysis, RNA-seq Data, differential gene expression, RNA Sequencing, RNA-seq differential analysis
1276	13	Breast Cancer, pathologic complete response, Neoadjuvant Chemotherapy, triple-negative breast cancer, pCR
1526	12	single-cell RNA sequencing, cell types, spatial transcriptomics, transcriptomic cell types, RNA sequencing data
22059	12	immune cell infiltration, Cancer Genome Atlas, immune cells, tumor immune microenvironment, breast cancer
1318	11	gene expression data, gene selection, Microarray gene expression, cancer classification, Feature selection methods
26279	11	Triple-negative breast cancer, breast cancer, TNBC cancer cells, TNBC patients, metastatic TNBC cancer
14251	9	cancer driver genes, Cancer Genome Atlas, cancer types, driver mutations, identify cancer driver

Top GitHub repositories

Repository	Number of within-cluster citations	Number of stars
BonilKoo/patient_stratification	1	0
XiaomeiLi1/CancerSubtypesPrognosis	1	1
rlupat/moanna	1	4
s-a-nersisyan/ExhauFS	1	10
nrohani/MolecularSubtypes	1	8
weiba/ERGCN	1	0
mehmetgonen/gsbc	1	5
amirfrz/BMDA	1	0
tanviralambd/Metabric	1	0
bbisr-shinyapps/NOJAH	1	2

The Map of Science runs on bibliometric data from [Clarivate Web of Science](#) and other sources. For details on our sourcing, methodology, and limitations, see the [tool documentation](#).